

Introductory Lecture: Methods Overview and Taxonomy

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Course Overview

- **Focus:** Uncertainty Quantification (UQ) in Machine Learning & Deep Learning, with engineering measurement aspects
- **Goal:** Understand the sources, methods, and applications of uncertainty estimation
- **Engineering angle:** the problems engineers actually face with ML
 - Measurement noise; small datasets; data that is *not* i.i.d.; genuine uncertainty
 - Bayesian view: learning from little data $\Rightarrow$ stronger modelling assumptions
- **Topics:**
 - Types of uncertainty (aleatoric vs. epistemic) and how to decompose them
 - Major UQ method categories (Bayesian, ensembles, single-pass, post-hoc, ...)
 - Advanced: physics-informed ML, generative models, state-space methods
 - Recent trends & open problems (conformal prediction, deterministic UQ, evaluation)
- **Practical:** implementation, calibration, and evaluation metrics

Why Uncertainty Quantification? The Core Problem

A modern neural network will confidently give you an answer even when it has no idea.

Standard deep networks are systematically overconfident:

- The softmax score is *not* a probability of being correct
- Accuracy improved over the last decade, but *calibration got worse*
- Overconfidence persists on out-of-distribution inputs

Why this is dangerous: a wrong-but-confident prediction fails *silently*.

- No error is raised, no human is alerted
- The system cannot know *when to ask for help*

What UQ buys us: the model can say “*I don't know*” → abstain, defer to a human, collect data, or trigger a safe fallback.

Why It Matters: Autonomous Driving

Perception, prediction and planning all consume model outputs that may be wrong.

Where uncertainty enters:

- **Sensors** (LiDAR, camera, radar): noise, glare, rain, fog → *aleatoric*
- **Out-of-distribution scenes**: unseen objects, novel environments → *epistemic*
- **Sensor fusion**: calibration / synchronization drift, dropped modality
- **Trajectory prediction**: other agents are inherently multi-modal

Why UQ is essential:

- If perception under-reports its own error, the planner leaves too little margin → collisions
- UQ surfaces rising uncertainty, which a runtime monitor can use as an early-warning trigger for a minimum-risk maneuver
- Increasingly required by safety standards (e.g. SOTIF / ISO 21448, ISO/PAS 8800 for AI)

Why It Matters: Computer Vision & Medical Imaging

In high-stakes vision, *knowing what you don't know* is as important as being accurate.

Computer vision:

- **Semantic segmentation:** per-pixel uncertainty flags unreliable regions (road/obstacle boundaries)
- **OOD / anomaly detection:** reject inputs the model was never trained on
- **Active learning:** label the most uncertain images first → cut annotation cost

Medical imaging & diagnosis:

- **Referral / abstention:** send low-confidence cases to a clinician
- **Distribution shift:** new scanner, hospital, or population breaks silently without UQ

Why It Matters: Engineering, Sensing & Forecasting

In engineering, a measurement without an error bar is not a measurement.

- **Metrology / measurement:** reporting a value *requires* an uncertainty budget (GUM)
- **Robotics & navigation:** state estimation is uncertainty propagation (Kalman/particle filters)
- **Time-series & forecasting:** energy, finance, health monitoring need calibrated intervals, not point predictions
- **Scientific ML:** sparse, expensive data; physics constraints; inverse problems
- **Predictive maintenance / anomaly detection:** act only when confidence is high enough

Common thread: the output feeds a **decision**, and decisions need risk, not just a guess.

Aleatoric Uncertainty (Noise)

Definition: Irreducible randomness inherent in the data and measurement process.

Sources:

① Noise in Data / Measurement Error

- Sensor noise (LIDAR, camera measurement errors)
- Environmental variations (weather, lighting)
- Cannot be reduced with more data

② Label Ambiguity / Labeling Noise

- Subjective labeling (e.g., medical diagnosis disagreement)
- Multiple valid answers for same input

③ Missing Information / Occlusion

- Occluded/hidden objects, unmeasured variables
- Incomplete features or temporal history

Key Property: *Irreducible* - persists even with infinite data

Epistemic Uncertainty (Model Uncertainty)

Definition: Reducible uncertainty due to lack of knowledge or incomplete model specification.

Sources:

- 1 **Limited Training Data / Small Dataset:** few samples with high-dimensional inputs; data sparsity in input space
- 2 **Wrong or Incomplete Model Architecture:** underspecified (too simple); missing interactions or nonlinearity
- 3 **Parameters Cannot be Learned from Data:** non-identifiability (many weight configs fit equally well); flat likelihood does not constrain all parameters
- 4 **Distribution Shift / Domain Shift:** covariate shift, out-of-distribution inputs; domain adaptation challenges

Key Property: *Reducible* - decreases with more data or better model

Uncertainty Decomposition and Implication

$$\text{Total Uncertainty} = \text{Aleatoric} + \text{Epistemic}$$

Epistemic	Aleatoric	Interpretation	Action
High	Low	Need more data	Collect data / Active learning
Low	High	Task inherently ambiguous	Set safety margins / Wider intervals
Low	Low	High confidence	Proceed with confidence
High	High	Unknown unknowns	Reject or request human review

Practical Decision Making:

- **High Epistemic** → *Slow down, collect more data*
- **High Aleatoric** → *Predict well, set wider intervals*
- **Both Low** → *Confidence, proceed*
- **Both High** → *Reject, human intervention*

Example 1: Radar Ranging (Conceptual)

A radar estimates the distance to a *stationary* object.

A simple measurement model:

$$z = r + b + \varepsilon$$

- r : true range b : unknown (fixed) calibration offset ε : random measurement noise

Aleatoric = variability from ε . Repeated readings fluctuate even with a perfectly known, correctly calibrated model. *Irreducible*.

Epistemic = uncertainty about the offset b (radar not yet tested enough). Measuring at *known* distances estimates b and reduces it. *Reducible*.

Key point: once b is accurately estimated, the noise ε still remains.

Caveat: this assumes b is stationary — if the offset drifts (temperature, ageing), part of it becomes an irreducible, time-varying term.

Major Categories of UQ Methods

Inference (Bayesian Methods)

- Bayesian Linear Regression, Bayesian Neural Networks, Gaussian Processes

Sampling & Ensemble Methods (Multi-Pass)

- Deep Ensembles, MC Dropout, Bootstrap, Snapshot Ensembles

Direct / Single-Pass Deterministic

- Heteroscedastic Regression, Quantile Regression, Evidential Deep Learning

Post-hoc Methods (No Model Change)

- Conformal Prediction, Temperature Scaling, Isotonic Regression

Generative Models

- VAE, Normalizing Flows, Diffusion Models

State-Space & Sequential

- Kalman Filter, Particle Filter, Hidden Markov Models

Physics-Informed ML

- Physics-Informed Neural Networks (PINNs), Neural Operators

Sequential Decision Making

- Reinforcement learning

Bayesian Linear Regression

Type: Epistemic Uncertainty

Core Idea:

- Place a prior distribution over weights: $p(\mathbf{w})$
- Compute posterior given data: $p(\mathbf{w}|\mathcal{D})$
- Output: Predictive distribution $p(y|\mathbf{x}, \mathcal{D})$

Advantages:

- Closed-form posterior (Gaussian)
- Interpretable weight uncertainty
- Uncertainty decreases with data

Disadvantages:

- Limited to linear models
- Does not scale to deep networks

Output: Mean prediction + variance (credible intervals)

Bayesian Neural Networks (BNN)

Type: Epistemic Uncertainty

Core Idea:

- Place prior over ALL weights $p(\mathbf{W})$
- Approximate posterior via Variational Inference (VI) or MCMC
- Sample from posterior at test time: $\hat{y} \sim \int p(y|\mathbf{x}, \mathbf{W})p(\mathbf{W}|\mathcal{D})d\mathbf{W}$

Advantages:

- Principled uncertainty quantification
- Works with deep networks
- Clear epistemic/aleatoric separation

Disadvantages:

- Computationally expensive (requires VI or MCMC)
- Posterior approximation quality varies

Output: Distribution over predictions

Gaussian Processes (GP)

Type: Epistemic Uncertainty

Core Idea:

- Non-parametric Bayesian method
- Define distribution over functions via kernel
- Posterior: $p(f|\mathcal{D}) \sim \mathcal{GP}(\mu(x), k(x, x'))$

Advantages:

- Exact posterior (no approximation)
- Principled uncertainty quantification
- Closed-form predictions with variance

Disadvantages:

- Scales as $O(n^3)$ in data size
- Limited to relatively small datasets
- Difficult to extend to high dimensions

Output: Mean + variance predictions

Deep Ensembles

Type: Epistemic Uncertainty

Core Idea:

- Train M independent models with different random initializations [2]
- At test time: get predictions from all M models
- Uncertainty from disagreement: $\text{Var}[\hat{y}] = \frac{1}{M} \sum_i (\hat{y}_i - \bar{y})^2$

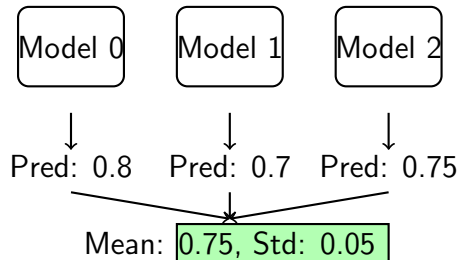
Advantages:

- Simple, embarrassingly parallel
- Gold standard for predictive uncertainty
- No model modification required
- Works with any architecture

Disadvantages:

- $M \times$ computational cost
- Requires multiple training runs

Output: Mean + variance of M predictions



Monte Carlo Dropout

Type: Epistemic Uncertainty

Core Idea:

- Keep dropout active at test time (not just training)
- Perform T stochastic forward passes
- Uncertainty from prediction variance across T passes

Key Insight: Dropout approximates Bayesian Neural Network [1]

Advantages:

- Single training run
- Minimal code changes (just enable dropout at test)

Disadvantages:

- Requires T forward passes at test (costly)
- Quality depends on dropout placement

Output: Mean + variance from T stochastic passes

Heteroscedastic Regression

Type: Aleatoric Uncertainty

Core Idea:

- Network has TWO output heads: $\mu(\mathbf{x})$ and $\log \sigma^2(\mathbf{x})$
- Learn input-dependent noise level
- Loss: Gaussian NLL = $\frac{1}{2} \exp(-\log \sigma^2)(y - \mu)^2 + \frac{1}{2} \log \sigma^2$

Advantages:

- Single forward pass
- Captures input-dependent noise
- Simple to implement

Disadvantages:

- Only captures aleatoric uncertainty
- Assumes Gaussian output noise
- Does not capture epistemic uncertainty

Output: Mean + predicted variance $\sigma^2(\mathbf{x})$

Quantile Regression

Type: Aleatoric Uncertainty

Core Idea:

- Predict multiple quantiles $\{q_\alpha, q_{0.5}, q_{1-\alpha}\}$
- Use pinball loss for quantile τ : $L_\tau = \max(\tau(y - \hat{y}), (1 - \tau)(\hat{y} - y))$
- Output: Prediction intervals $[q_\alpha, q_{1-\alpha}]$

Advantages:

- Distribution-free (no Gaussian assumption)
- Single forward pass
- Flexible interval widths

Disadvantages:

- Only captures aleatoric uncertainty
- Need to train separate heads per quantile
- Quantiles may not be monotonic

Output: Prediction intervals

Evidential Deep Learning (EDL) – With Caveats

Type: Aims to separate Epistemic + Aleatoric in ONE pass

Core idea: predict the parameters of a higher-order (Dirichlet) distribution; $S = \sum_k \alpha_k$ is the “evidence”. Aleatoric = entropy of the mean; epistemic $\propto 1/S$. [3]

Attractive because: single forward pass, strong empirical OOD detection, one model.

But recent work urges caution (important for this course):

- The learned *epistemic* uncertainty can be unreliable — e.g. it need not vanish even with infinite data [15]
- EDL is often better understood as an *energy-based OOD detector* than as true epistemic UQ [16]
- Its uncertainty is largely controlled by the regularizer $\rightarrow$ meaningful only in a *relative* sense

Bottom line: useful and fast, but don't treat its epistemic output as calibrated ground truth — validate against ensembles/conformal on your task.

Conformal Prediction

Type: Neither (Distribution-Free Coverage)

Core Idea:

- Wrapper around ANY pre-trained predictor [4]
- Split calibration data by non-conformity score

Advantages:

- Model-agnostic
- Finite-sample coverage guarantee
- NO retraining required
- Valid regardless of model quality

Disadvantages:

- Requires held-out calibration data
- Intervals can be conservative
- Exchangeability assumption

Output: Prediction sets/intervals with guaranteed coverage

Diffusion Models (DDPM)

Type: Both (Epistemic in denoising, Aleatoric output) — **Generative:** YES

Core Idea:

- **Forward process:** Add Gaussian noise iteratively $q(x_t|x_{t-1})$
- **Reverse process:** Learn denoiser $p_\theta(x_{t-1}|x_t)$ via score matching
- **Sampling:** Iterate from $x_T \sim \mathcal{N}(0, I)$ down to x_0 [5]

Uncertainty Quantification:

- Generate M samples from $p(x|y)$
- Compute variance across samples
- Score network variance as epistemic uncertainty

Advantages:

- State-of-the-art generation quality
- Flexible conditioning
- Sample diversity = natural uncertainty

Disadvantages:

- Computationally expensive (many reverse steps)
- Long sampling time

Output: Samples from $p(x|y)$ with inherent diversity

Kalman Filter / EKF

Type: Both (Process noise = Aleatoric, State = Epistemic)

Core Idea:

- Linear-Gaussian state-space model:

$$\mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q})$$

$$\mathbf{y}_t = \mathbf{H}\mathbf{x}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(0, \mathbf{R})$$

- Recursive Bayesian filter: Prediction $\rightarrow$ Update

Output: Posterior $p(\mathbf{x}_t | \mathbf{y}_{1:t})$ with covariance $\mathbf{P}_t$

Uncertainty:

- **Aleatoric:** Process noise $\mathbf{Q}$, measurement noise $\mathbf{R}$
- **Epistemic:** State covariance $\mathbf{P}_t$ (shrinks with observations)

Applications: Navigation, tracking, time-series forecasting

Physics-Informed Neural Networks (PINNs)

Type: Epistemic (Parameter + Model Uncertainty)

Core Idea:

- Incorporate partial differential equation as a constraint in loss:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}}$$

where $\mathcal{L}_{\text{physics}}$ is a loss function that includes physical model

- Network learns to satisfy both data and physics simultaneously

Advantages:

- Learn from sparse data (PDE provides regularization)
- Physically consistent solutions
- Inverse problem capability

Applications: Fluid dynamics, heat transfer, inverse problems

What the Field Is Working On Now (2023–2026)

Distribution-free UQ has exploded

- Conformal prediction is now a research area of its own (see next slides)

Cheap, single-pass uncertainty

- Deterministic distance-aware methods (SNGP, DUQ, DDU); last-layer Laplace
- Efficient ensembles: BatchEnsemble, MIMO, Packed-Ensembles

A skeptical turn: does the aleatoric/epistemic split actually work?

- Benchmarks show the two components are often *entangled* in practice
- Evidential deep learning's epistemic estimates are being questioned

Better evaluation

- Critiques of ECE; proper scoring rules; disentanglement / OOD benchmarks
- Open toolkits: Torch-Uncertainty, Lightning-UQ-Box, Laplace, Fortuna

Uncertainty for generative & scientific ML

- Diffusion models as samplers of predictive distributions; Bayesian PINNs / neural operators

How Do We Know an Uncertainty Is *Good*?

Evaluation is a first-class topic — “it has error bars” is not enough.

Calibration — do predicted probabilities match observed frequencies?

- Reliability diagrams; **Expected Calibration Error (ECE)** — but ECE is biased, binning-dependent, and can be gamed
- Fixes: post-hoc *temperature scaling*; adaptive binning

Proper scoring rules — reward calibration *and* sharpness together

- NLL, Brier score (classification); CRPS (regression/probabilistic forecasts)

Task-specific evaluation

- Intervals: empirical *coverage* vs. nominal, and *width* (sharpness)
- Epistemic: OOD-detection AUROC; selective prediction / accuracy-vs-rejection curves

Rule of thumb: report calibration *and* sharpness — a model that always says “50%” is calibrated but useless.

Practical Tooling & Benchmarks

So you don't reimplement everything from scratch in the labs.

Libraries

- **Torch-Uncertainty**, **Lightning-UQ-Box** — ensembles, MC dropout, packed ensembles, conformal, metrics
- **Laplace** (laplace-torch) — last-layer / full Laplace in a few lines
- **MAPIE**, **TorchCP** — conformal prediction; **Fortuna** — calibration & conformal
- **Uncertainty Baselines** (Google) — reference implementations

Benchmarks / shifts to test on

- CIFAR-10/100-C, ImageNet-C (corruptions); OOD pairs (CIFAR vs. SVHN)
- WILDS (real-world distribution shift); disentanglement benchmarks

Summary: 7 UQ Method Categories

Category	Methods	Single-Pass?	Uncertain Type	Multi-Pass
1. Inference	BNN, GP, Bayes LR	Partial	Epistemic	Yes (MCMC)
2. Sampling/Ensemble	Deep Ens, MC Drop, Bootstrap	No	Epistemic	Multiple
3. Direct Single-Pass	Hetero, Quantile, EDL	YES	Aleatoric/Both	Single
4. Post-hoc	Conformal, Temp Scale	YES	Neither	Single
5. Generative	VAE, Diffusion, Flows	Partial	Both	Multiple (sampling)
6. State-Space	Kalman, Particle, HMM	N/A	Both	Sequential
7. Physics-Informed	PINNs, Operators	No	Epistemic	Ensemble

Cross-cutting (Part 3): distance-aware single-pass (SNGP/DUQ/DDU), efficient ensembles & Laplace, modern conformal, and *evaluation* (calibration + sharpness) apply across all seven.

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